

Unity VR and AR Setup Guide

1. Create the Unity Project

1. Open **Unity Hub**.
2. Create a **New Project**.
3. Select the **Universal 3D (URP)** template.
4. Name the project and choose a location.
5. Click **Create Project**.

Part 1 – VR Setup (Meta Quest)

2. Switch Build Platform to Android

1. Open **File** → **Build Settings**.
2. Select **Android** from the platform list.
3. Click **Switch Platform**.

3. Create Basic Scene Objects

1. In the **Hierarchy**, create a **Capsule**
 - a. Adjust the **Transform Position Y value** to **1**.
2. Create a **Plane**
 - a. Set **Scale** to **(10, 10, 10)**.

These objects act as a simple environment for VR testing.

4. Configure Input Settings

1. Open **Edit → Project Settings → Player**.
2. Under **Active Input Handling**, select **Input System Package (New)**.
3. When prompted, allow Unity to **restart the editor**.

5. Configure Package Identifier

1. In **Project Settings → Player → Identification**:
2. Uncheck **Override Default Package Name**, then recheck it.
3. Enter a unique **Package Name** (e.g., `com.company.projectname`).

6. Install XR Plugin Management

1. Go to **Edit → Project Settings → XR Plugin Management**.
2. Click **Install XR Plugin Management** if it is not already installed.
3. Select the **Android** tab.
4. Enable **OpenXR**.

7. Install Meta XR SDK

1. Open **Window → Package Manager**.
2. Install **Meta XR All-in-One SDK**.

If it is not available:

- Open the **Asset Store**
- Search for **Meta XR All-in-One SDK**
- Download and import it into the project.

8. Add Meta XR Building Blocks

After installing the SDK:

1. In the **top menu bar**, select

Meta → Meta XR Tools → Building Blocks.

2. Add the following components to your scene by **dragging them into the Hierarchy**:
 - a. **Camera Rig**
 - b. **Controller Tracking**
 - c. **Virtual Hands**

9. Configure OpenXR Interaction Profiles

1. Open **Edit → Project Settings → XR Plugin Management → OpenXR.**
2. Under **Enabled Interaction Profiles**, add:
 - a. **Meta Quest Touch Plus Controller Profile**

10. Enable Meta Quest Support Features

Ensure the following features are enabled:

- **Meta Quest Support**
- **Meta XR Features**
- **Meta XR Foundation**

Next:

1. Click the **Settings icon** next to **Meta Quest Support**.
2. Ensure **all Quest devices** are selected.

11. Build and Run for VR

1. Connect a **Meta Quest headset** with developer mode enabled.
2. Open **File → Build Settings**.
3. Click **Build and Run**.

Part 2 – AR Setup (ARCore)

This section configures the same project for **Android AR using ARCore**.

12. Install AR Foundation

1. Open **Window → Package Manager**.
2. Select **Unity Registry**.
3. Install:
 - a. **AR Foundation**

13. Disable VR Cameras

Since the project will now use AR:

1. In the **Hierarchy**, disable:
 - a. **VR Camera Rig**
 - b. **Main Camera**

14. Add AR Components

In the **Hierarchy**:

1. Click **+** → **XR** → **XR Origin (Mobile AR)**
 - a. This creates the AR camera system.
2. Click **+** → **XR** → **AR Session**
 - a. This component manages the AR lifecycle.

15. Configure AR Background Rendering

1. Navigate to **Assets** → **Settings**.
2. Select the **Mobile Renderer** asset.
3. In the **Inspector**, click **Add Renderer Feature**.
4. Select **AR Background Renderer Feature**.

16. Switch XR Plugin to ARCore

1. Go to **Edit** → **Project Settings** → **XR Plugin Management**.
2. **Disable OpenXR**.
3. **Enable ARCore**.

17. Adjust Graphics API

1. Go to **Edit** → **Project Settings** → **Player** → **Other Settings**.
2. Under **Graphics APIs**, remove **Vulkan**.
3. Ensure **OpenGL ES3** is enabled.

18. Build and Run for AR

1. Connect an **ARCore-compatible Android device** (e.g., Pixel phone).
2. Open **File** → **Build Settings**.
3. Click **Build and Run**.

✓ After completing these steps, the project should support:

- **VR deployment on Meta Quest devices**
- **AR deployment on ARCore-enabled Android phones**

If you want, I can also:

- Turn this into a **clean internal studio SOP document**
- Add **recommended Unity versions and package versions**
- Add **common errors and fixes for Quest builds** (these save hours during setup).